REFLEX: Reflective Evolution from LLM Experience

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HarnessRSI 71 ★★★

evidence: retained

Paper

1 What changes, what is kept

Evolved object: a program (policy code / design function)
model weights never change · train-free

1 Run the parent
fitness f + a Behavioral Evidence image (trajectory, phase portrait, array cut)

2 Vision Critic diagnoses
image + f in; JSON out: failure mode, root cause, fix. It writes no code

3 BM25 retrieves skills
diagnosis = query; top 3 code snippets from the Skill Memory (≤ 256)

4 Text-only Actor writes child
parent code + diagnosis + snippets; the Actor never sees the image

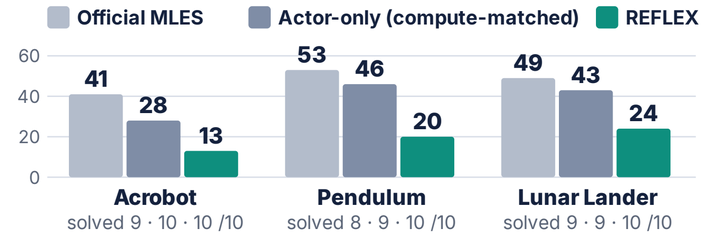
5 Evaluate, keep, credit
keep 16 best; used snippets get Δ = f(child) − f(parent); Δ > 0 adds a new snippet

loop

UCB1 picks 1 of 5 operators each step
archived if utility < −1 after ≥ 3 retrievals
every Critic, Actor, repair, failed call counts

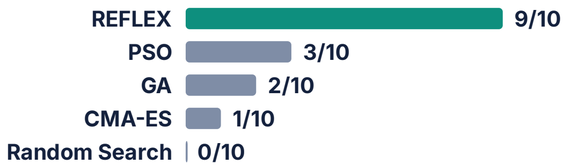
2 Headline: about half the LLM calls to solve

Median LLM calls to solve · 100 calls, 10 paired seeds (Table 1)



Same weights; every attempt counted. Solved: MLES · Actor-only · REFLEX.

36-dim antenna design · seeds reaching score 25.25 (Table 2)



240 evals, shared first designs for all methods; LLM cost not matched.

3 Controls that decide if the gain is real

Is the image what matters? (Table 3a)

Critic input (Acrobot)	AUC	calls to target
correct image (full REFLEX)	**−0.2850**	**5.50**
Actor-only (no Critic)	−0.3450	10.30
blank image	−0.3620	9.50
shuffled image	**−0.4270**	**14.70**

AUC gain +0.142 / +0.077 / +0.060 vs shuffled / blank / Actor-only (p = 0.004 / 0.012 / 0.012); final score p = 0.18. **Acrobot only, mean ± SEM, no test on calls; budget unstated, not reconcilable with Table 1.**

Frozen Skill Memory transfer (Sec. 4.4)

Acrobot, 20 calls	final	AUC@20	calls
empty bank	−0.139	−0.523	4.0
Pendulum top-2	**−0.105**	−0.310	2.0
p (final · AUC · calls)	0.002	**0.049**	**0.039**
Lunar Lander top-2	**−0.118**	−0.380	2.5
p (final · AUC · calls)	0.002	**0.065**	**0.91**

10/10 paired seeds per bank; banks hash-verified and audited. **Target is −0.20: cold mean is already above it.** One target task, one model; the Critic shows no benefit at this 20-call cap.

Matched: equal total LLM calls; 40% of REFLEX calls go to the Critic (27 vs 48 programs evaluated). Actor-only keeps the Skill Memory rules, but its placeholder diagnosis is also the BM25 query.

Memory object (Table 3b): executable vs textual p = 0.82; AUC@20 −0.5445 vs −0.5618 with no memory; source bank not stated.

4 Where the gain is absent or thin

- **Lunar Lander final score**, 100 calls: 1.082 REFLEX, 1.064 Actor-only, 1.069 MLES
- **20-call cap, Acrobot:** REFLEX and Actor-only both solve 6/10 (final-score p = 0.92)
- **Transfer:** Lunar Lander bank p = 0.065 (AUC), 0.91 (calls); 2 source–target pairs, 1 model
- **Antenna:** GA, PSO also reach 25.0 on 10/10; baseline tuning, tests not reported

Takeaway

A vision Critic roughly halves the LLM calls needed to solve three control tasks, versus a compute-matched Actor-only run; an Acrobot image-shuffle control points to the image. Final scores are close, 20 calls are too few, and Skill Memory transfer rests on one target task.

Our read · HarnessRSI

Score 71 · ★★★ · evidence level: retained

Reports: matched calls, paired seeds, image shuffle, transfer.
Not reported: Skill Memory ablation (main runs), improver comparison, code.